

Name \_\_\_\_\_ Teacher \_\_\_\_\_



# **GOSFORD HIGH SCHOOL**

**2012**

## **HIGHER SCHOOL CERTIFICATE**

### **ASSESSMENT TASK 2**

# **MATHEMATICS**

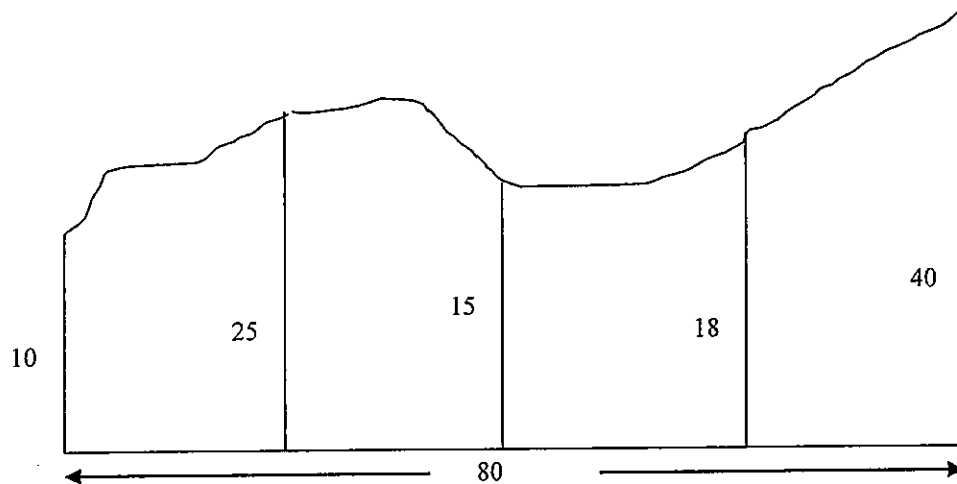
**Duration-** 90 minutes plus 5 minutes reading time

|                                                                                                                                   |            |
|-----------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Multiple choice</b> 6 questions worth <b>1 mark each</b> . (Write your answer on your own paper, <b>not</b> on the test paper) | <b>/6</b>  |
| <b>Applications of differentiation.</b>                                                                                           | <b>/22</b> |
| <b>Integration</b>                                                                                                                | <b>/32</b> |
| <b>TOTAL</b>                                                                                                                      | <b>/60</b> |

### Part A

**Multiple choice.** Write your answers on your own paper not on the question sheet.

#### Question 1.



The field drawn is to have its area approximated by applying Simpson's Rule. The value of  $h$  is:

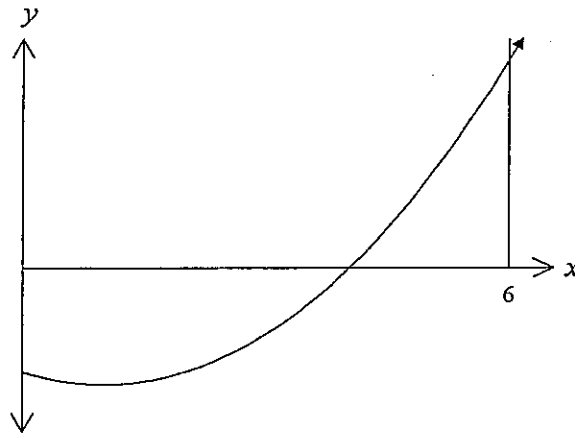
- (A) 80      (B) 40      (C) 20      (D) 10

#### Question 2.

For what values of  $x$  is the curve  $f(x) = 2x^3 + x^2$  concave down?

- (A)  $x < -\frac{1}{6}$       (B)  $x > -\frac{1}{6}$   
(C)  $x < -6$       (D)  $x > 6$

Question 3.

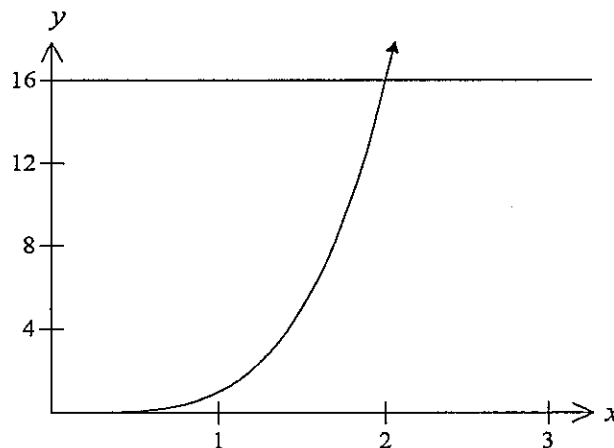


What is the correct expression for the area bounded by the  $x$ -axis and the curve  $y = x^2 - 2x - 8$  between  $0 \leq x \leq 6$ ?

- (A)  $A = \int_0^5 x^2 - 2x - 8 dx + \left| \int_5^6 x^2 - 2x - 8 dx \right|$
- (B)  $A = \int_0^4 x^2 - 2x - 8 dx + \left| \int_4^6 x^2 - 2x - 8 dx \right|$
- (C)  $A = \left| \int_0^5 x^2 - 2x - 8 dx \right| + \int_5^6 x^2 - 2x - 8 dx$
- (D)  $A = \left| \int_0^4 x^2 - 2x - 8 dx \right| + \int_4^6 x^2 - 2x - 8 dx$

Question 4.

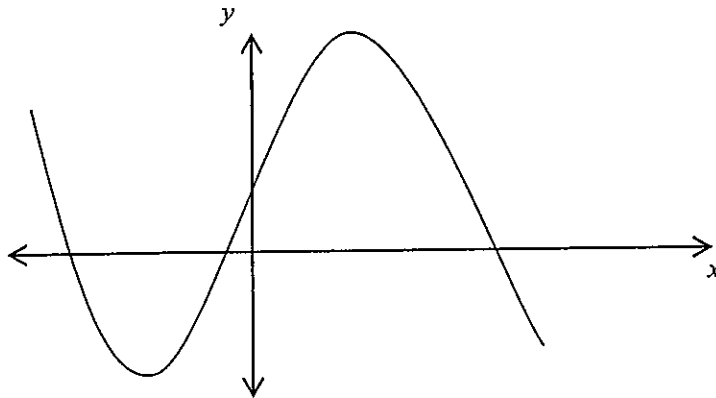
A region in the diagram is bounded by the curve  $y = x^4$ , the  $y$ -axis and the line  $y = 16$ .



Which of the following expressions is correct for the volume of the solid of revolution when this region is rotated about the  $y$ -axis?

- (A)  $V = \pi \int_0^2 x^8 dx$
- (B)  $V = \pi \int_0^{16} x^8 dx$
- (C)  $V = \pi \int_0^2 y^{\frac{1}{4}} dy$
- (D)  $V = \pi \int_0^{16} y^{\frac{1}{4}} dy$

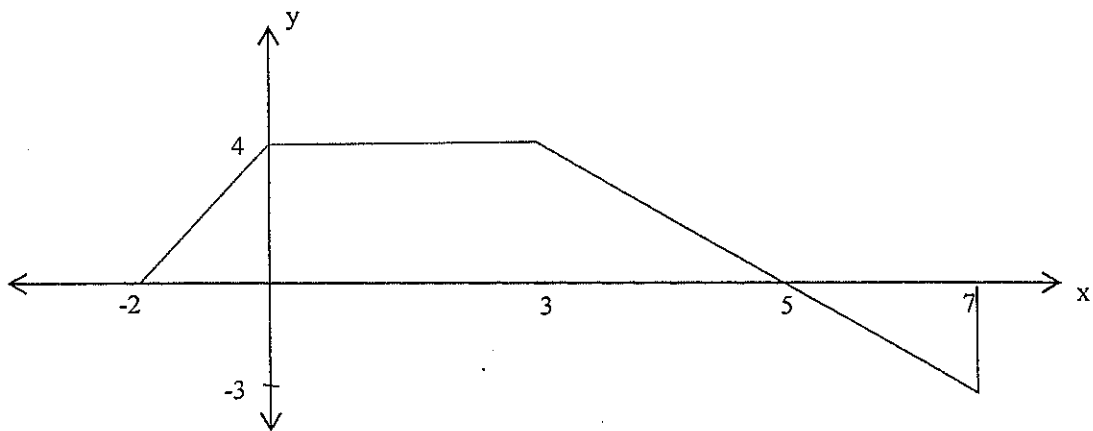
**Question 5**



The graph shows  $y = f'(x)$ . The graph of  $y = f(x)$  has:

- (A) A maximum and a minimum turning point.
- (B) Two maximum and a minimum turning point
- (C) Two minimum and a maximum turning point.
- (D) Two maximum turning points and a horizontal point of inflection.

**Question 6**



The graph of  $y = f(x)$  is given above. The value of  $\int_{-2}^7 f(x) dx$  is:

- (A) 23
- (B) 17
- (C) 19
- (D) 13

## PART B

### Question 1.

a) Find the following indefinite integrals.

i)  $\int x^3 + 6x - 1 \, dx$  (1)

ii)  $\int (3x - 2)^5 \, dx$  (1)

iii)  $\int \frac{x^3 + 5x^2 - 1}{x^2} \, dx$  (2)

iv)  $\int x\sqrt{x} \, dx$  (2)

b) Evaluate

i)  $\int_1^3 6x^2 - 1 \, dx$  (2)

ii)  $\int_{-3}^3 \sqrt{9 - x^2} \, dx$  (1)

### Question 2. (Start a new page)

a) A continuous curve  $y = f(x)$  has the following properties over the domain  $0 \leq x \leq 6$ :  $f(x) \geq 0$ ,  $f'(x) \geq 0$  and  $f''(x) \geq 0$ . Sketch a curve satisfying these conditions over the given domain. (1)

b) If  $\frac{d^2y}{dx^2} = 12x + 6$  and  $\frac{dy}{dx} = 1$  at the point  $(-1, -2)$  find the equation of the curve. (3)

c) The table shows the values of  $x \log_e x$  for 5 values of  $x$ . By using Simpson's rule find an approximation (to 1 decimal place) for the area between the curve  $y = x \log_e x$  and the  $x$  axis from  $x = 1$  to  $x = 5$ . (2)

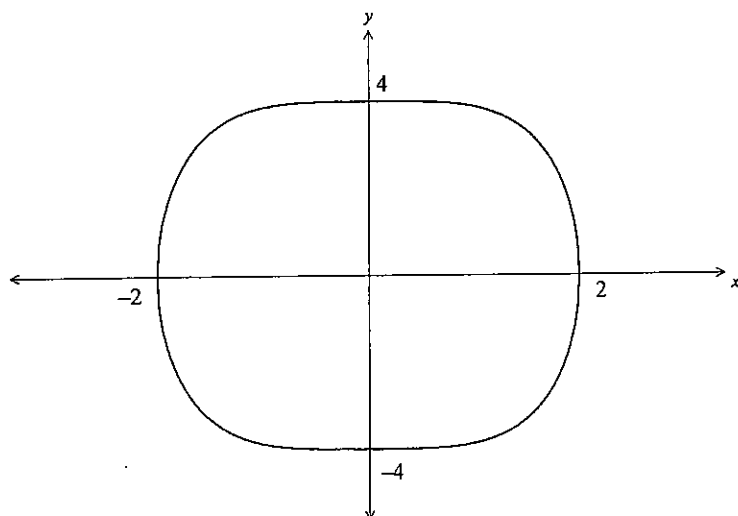
|              |   |     |     |     |     |
|--------------|---|-----|-----|-----|-----|
| $x$          | 1 | 2   | 3   | 4   | 5   |
| $x \log_e x$ | 0 | 1.4 | 3.3 | 5.5 | 8.1 |

d) i) Find the points of intersection for the curves  $y = x^2$  and  $x = y^2$  (1)

ii) Hence find the area between the two curves. (2)

**Question 3** (Start a new page)

- a) The region bounded by the  $x$  axis and the curve  $y = \frac{1}{x^2}$  from  $x = 1$  to  $x = b$  is  $\frac{1}{2}$  a square unit. What is the value of  $b$ ? (2)
- b) Find the area enclosed between the curve  $y = x(1 - x)$  and the  $x$  axis from  $x = 0$  to  $x = 2$  (3)
- c) The sketch shows the curve  $y^2 + x^4 = 16$ .



The area enclosed within the curve is rotated about the  $x$  axis.  
Find the volume of the solid of revolution so formed. (2)

- d) By using calculus find the volume of the solid generated by rotating the line  $2x + y = 1$  about the  $y$  axis from  $y = -1$  to  $y = 1$ . (2)

**Question 4** (Start a new page)

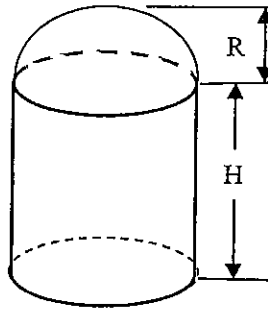
- a) Use the trapezoidal rule with 5 function values to approximate  $\int_0^1 \sqrt{1 - x^2} dx$  (give your answer correct to 2 decimal places) (3)
- b)
- i) On the same number plane sketch the curves  $y = 2 - x^2$  and  $y = x^2$ . (2)
- ii) Given that the two curves intersect at  $(1,1)$  in the first quadrant calculate the area of the region, **in the first quadrant**, between the two curves and the  $x$  axis. (4)  
(correct to 3 decimal places)

**Question 5.** (Start a new page)

Consider the curve given by  $y = 1 + 3x - x^3$  for  $-2 \leq x \leq 3$ .

- a) Find the turning points and determine their nature. (3)
- b) What are the coordinates of any points of inflexion? (2)
- c) Sketch the curve for  $-2 \leq x \leq 3$ . (2)
- d) What are the maximum and minimum values of  $y$  for  $-2 \leq x \leq 3$ . (2)

**Question 6.** (Start a new page)



The diagram illustrates the shape of a new wheat silo, comprising a hemisphere on top of an open cylinder. It is to be erected on an existing concrete slab, which acts as the base of the silo. The volume is to be  $576\pi$  cubic metres. The builder charges \$10 per square metre to build the cylinder and \$20 per square metre to build the hemisphere.

- i) Given that the radius of the hemisphere is  $R$  metres and the height of the cylinder is  $H$  metres write down an expression for the volume of the silo in terms of  $H$  and  $R$ .  
(you may use the formulae  $V = \pi r^2 h, V = \frac{4}{3} \pi r^3$ ) (1)
- ii) Show that the cost of construction,  $P$  dollars, is given by  $P = 20\pi R(2R + H)$ .  
(you may use the formulae  $A = 4\pi r^2, A = 2\pi rh$ ) (2)
- iii) Hence prove that  $P = \frac{80\pi R^2}{3} + \frac{11520\pi}{R}$ . (3)
- iv) Hence find the minimum cost of construction, to the nearest dollar. (3)

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x$ ,  $x > 0$

PART A

- 1) C 2) A 3) D  
4) D 5) B 6) B

PART B

Q1) a) i)  $\int x^3 + 6x - 1 \, dx$   
 $= \frac{x^4}{4} + 3x^2 - x + C$

ii)  $\int (3x-2)^5 \, dx$   
 $= \frac{(3x-2)^6}{18} + C$

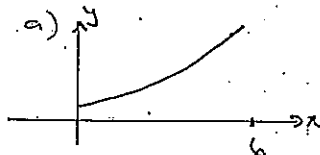
iii)  $\int \frac{x^3 + 5x^2 - 1}{x^2} \, dx$   
 $= \int x + 5 - \frac{1}{x^2} \, dx$   
 $= \frac{1}{2}x^2 + 5x + \frac{1}{x} + C$

iv)  $\int x\sqrt{x} \, dx$   
 $= \int x^{3/2} \, dx$   
 $= \frac{2x^{5/2}}{5}$   
 $= \frac{2\sqrt{x^5}}{5} + C$

b)  $\int_1^3 6x^2 - 1 \, dx$   
 $= \left[ 2x^3 - x \right]_1^3$   
 $= (54 - 3) - (2 - 1)$   
 $= 50$

ii)  $\int_{-3}^3 \sqrt{9-x^2} \, dx$   
 $= \frac{1}{2} \pi \times 3^2$   
 $= \frac{9\pi}{2}$

Q2)



b)  $\frac{d^2y}{dx^2} = 12x + 6$

$\frac{dy}{dx} = 6x^2 + 6x + C$

$x=1, \frac{dy}{dx} = 1$   
 $1 = 6 - 6 + C$   
 $\therefore C = 1$

$\frac{dy}{dx} = 6x^2 + 6x + 1$

$y = 2x^3 + 3x^2 + x + C$

passes through  $(-1, -2)$

$-2 = -2 + 3 - 1 + C$

$-2 = C$

$\therefore y = 2x^3 + 3x^2 + x - 2$

c)

$A = \frac{1}{3} \{1^3 + 1^3 + 2 \times 1 \times 1 + 4 \times 1 \times 1\}$   
 $= \frac{1}{3} \{0 + 8 + 1 + 2 \times 3 \times 3 + 4(1 \times 4 \times 5)\}$   
 $= \frac{1}{3} \{8 + 1 + 6 + 6 + 27 + 6\}$   
 $= \frac{1}{3} \times 42$   
 $= 14$

d)  $y = x^2 \dots (1)$   
 $x = y^2 \dots (2)$

Sub (2) into (1)

$y = y^4$

$y^4 - y = 0$

$y(y^3 - 1) = 0$

$y = 0, 1$

$\therefore x = 0, 1$

$\therefore$  pts.  $(0,0)$   $(1,1)$

ii)  $A = \int_0^1 \sqrt{x-x^2} \, dx$   
 $= \left[ \frac{2x^{3/2}}{3} - \frac{x^3}{3} \right]_0^1$

$= \left( \frac{2}{3} - \frac{1}{3} \right) - 0$

$= \frac{1}{3}$  Square units

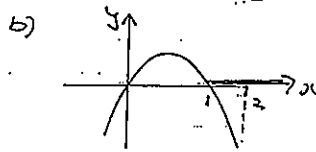
Q3) a)  $\int_1^b \frac{1}{x^2} \, dx = \frac{1}{2}$

$\left[ -\frac{1}{x} \right]_1^b = \frac{1}{2}$

$-\frac{1}{b} - (-1) = \frac{1}{2}$

$-\frac{1}{b} = -\frac{1}{2}$

$b = 2$



$A = \int_0^1 x - x^2 \, dx + \int_1^2 x - x^2 \, dx$   
 $= \left[ \frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 + \left[ \frac{x^2}{2} - \frac{x^3}{3} \right]_1^2$   
 $= \left( \frac{1}{2} - \frac{1}{3} \right) - 0 + \left( 2 - \frac{8}{3} \right) - \left( \frac{1}{2} - \frac{1}{3} \right)$   
 $= \frac{1}{6} + \left( -\frac{5}{6} \right)$   
 $= -\frac{2}{6} = -\frac{1}{3}$

c)  $\pi \int_{-2}^2 16 - x^4 \, dx$

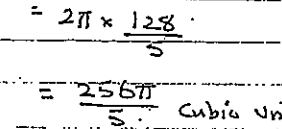
$= 2\pi \int_0^2 16 - x^4 \, dx$

$= 2\pi \left[ 16x - \frac{x^5}{5} \right]_0^2$

$= 2\pi \left[ (32 - \frac{32}{5}) - 0 \right]$

$= 2\pi \times \frac{128}{5}$

$= \frac{256\pi}{5}$  Cubic units



d)  $\pi \int_{-1}^1 \left( \frac{1-y}{2} \right)^2 \, dy$

$= \frac{\pi}{4} \int_{-1}^1 1 - 2y + y^2 \, dy$

$= \frac{\pi}{4} \left[ y - y^2 + \frac{y^3}{3} \right]_{-1}^1$

$= \frac{\pi}{4} \left[ \left( 1 - 1 + \frac{1}{3} \right) - \left( -1 - 1 - \frac{1}{3} \right) \right]$

$= \frac{\pi}{4} \left( \frac{1}{3} - (-2\frac{1}{3}) \right)$

$= \frac{\pi}{4} \left( \frac{8}{3} \right)$

$= \frac{2\pi}{3}$  Cubic units

Q4 a)  $\sqrt{1-x^2}$

|   |   |       |       |       |   |
|---|---|-------|-------|-------|---|
| x | 0 | 0.25  | 0.5   | 0.75  | 1 |
| y | 1 | 0.968 | 0.866 | 0.661 | 0 |

$A = \frac{h}{2} \{ \text{ends} + 2 \times \text{middle} \}$

$= 0.25 \{ 0 + 1 + 2(2.495) \}$

$= 0.125 \{ 5.99 \}$

$= 0.75$

b) i)  $y = 3 - 3x^2$

$y' = -6x$

turning pts.  $y' = 0$

$3 - 3x^2 = 0$

$3(1-x^2) = 0$

$3(1-x)(1+x) = 0$

$x = 1$  or  $x = -1$

$y = 3$   $y = -1$

$f''(1) < 0$   $f''(-1) > 0$

$\therefore (1,3)$  is max  $\therefore (-1,-1)$

b) possible point of inflection  $y'' = 0$

$-6x = 0$

$x = 0$

$y = 1$

$f''(-1) > 0$

$f''(1) < 0$

$\therefore$  change in concavity

$\therefore$  pt of inflection  $(0,1)$

c)  $x = -2$   $x = 3$

$y = 3$   $y = -1$

$A = \int_0^1 x^2 \, dx + \int_1^2 2 - x^2 \, dx$

$= \left[ \frac{x^3}{3} \right]_0^1 + \left[ 2x - \frac{x^3}{3} \right]_1^2$

$= \left( \frac{1}{3} - 0 \right) + \left( 2(2 - \frac{8}{3}) - (2 - \frac{1}{3}) \right)$

$= \frac{1}{3} + \frac{4}{3} - 1\frac{2}{3}$

$= 0.552$

Q5)

a)  $y = 1 + 3x - x^3$

$y' = 3 - 3x^2$

$y'' = -6x$

turning pts.  $y' = 0$

$3 - 3x^2 = 0$

$3(1-x^2) = 0$

$3(1-x)(1+x) = 0$

$x = 1$  or  $x = -1$

$y = 3$   $y = -1$

$f''(1) < 0$   $f''(-1) > 0$

$\therefore (1,3)$  is max  $\therefore (-1,-1)$

b) possible point of inflection  $y'' = 0$

$-6x = 0$

$x = 0$

$y = 1$

$f''(-1) > 0$

$f''(1) < 0$

$\therefore$  change in concavity

$\therefore$  pt of inflection  $(0,1)$

c)  $x = -2$   $x = 3$

$y = 3$   $y = -1$

$A = \int_0^1 x^2 \, dx + \int_1^2 2 - x^2 \, dx$

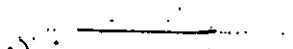
$= \left[ \frac{x^3}{3} \right]_0^1 + \left[ 2x - \frac{x^3}{3} \right]_1^2$

$= \left( \frac{1}{3} - 0 \right) + \left( 2(2 - \frac{8}{3}) - (2 - \frac{1}{3}) \right)$

$= \frac{1}{3} + \frac{4}{3} - 1\frac{2}{3}$

$= 0.552$

d) max 3, min -2



46)

$$i) V = \frac{1}{2} \left( \frac{4}{3} \pi r^3 \right) + \pi r^2 h$$

$$V = \frac{2}{3} \pi R^3 + \pi R^2 H$$

$$ii) A = \frac{1}{2} (4\pi R^2) + 2\pi R H$$

$$= 2\pi R^2 + 2\pi R H$$

$$\therefore P = 20 \times 2\pi R^2 + 10 \times 2\pi R H$$

$$= 20\pi R (2R + H)$$

iii) from (i)

$$576\pi = \frac{2}{3} \pi R^3 + \pi R^2 H$$

$$576 = \frac{2}{3} R^3 + R^2 H$$

$$1728 = 2R^3 + 3R^2 H$$

$$H = \frac{1728 - 2R^3}{3R^2}$$

Sub into (ii)

$$P = 20\pi R \left( 2R + \frac{1728 - 2R^3}{3R^2} \right)$$

$$= 20\pi R \left( \frac{6R^3 + 1728 - 2R^3}{3R^2} \right)$$

$$= 20\pi R \left( \frac{4R^3 + 1728}{3R^2} \right)$$

$$= 20\pi R \left( \frac{4R}{3} + \frac{1728}{3R^2} \right)$$

$$P = \frac{80\pi R^2}{3} + \frac{11520\pi}{R}$$

$$iv) P = \frac{80\pi R^2}{3} + \frac{11520\pi}{R}$$

$$\frac{dP}{dR} = \frac{160\pi R}{3} - \frac{11520\pi}{R^2}$$

$$\frac{d^2P}{dR^2} = \frac{160\pi}{3} + \frac{23040\pi}{R^3}$$

$$\min \frac{dP}{dR} = 0$$

$$\frac{160\pi R}{3} - \frac{11520\pi}{R^2} = 0$$

$$\frac{160\pi R}{3} = \frac{11520\pi}{R^2}$$

$$160\pi R^3 = 34560\pi$$

$$R^3 = 216$$

$$R = 6 \quad \left( \min \frac{d^2P}{dR^2} > 0 \text{ for all} \right)$$

$\therefore$  min cost when  $R = 6$

$$\therefore P = \frac{80\pi \times 6^2}{3} + \frac{11520\pi}{6}$$

$$= 960\pi + 1920\pi$$

$$= 2880\pi$$

$$= 9047.78$$

$$= \$9048$$